

Jayapoorani M

Data Analyst | Data Scientist | Python | SQL | Machine Learning | Power BI

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PROFESSIONAL SUMMARY

Data Analyst and aspiring Data Scientist with 5+ years of IT experience in data validation, SQL analytics, automation, and cloud data migration. Completed the Master Data Science Program at GUVI, with hands-on experience in Python, SQL, Machine Learning, Deep Learning, ETL, Power BI, Streamlit, and Data Visualization through multiple end-to-end projects. Skilled in analyzing enterprise-scale datasets, building predictive models, developing interactive dashboards, and delivering data-driven insights to support business decision-making.

SKILLS

Programming Languages: Python, SQL, Java, JavaScript

Data Analysis: Pandas, NumPy, Data Cleaning, EDA, Feature Engineering, Statistical Analysis, Data Validation, Data Quality

Machine Learning: Scikit-learn, Regression, Classification, Clustering, Random Forest, XGBoost, Decision Trees, KNN, SVM, Model Evaluation, Cross Validation, Hyperparameter Tuning

Deep Learning: TensorFlow, Keras, ANN, CNN, YOLOv8

Data Visualization: Power BI, Streamlit, Matplotlib, Seaborn

Databases: MySQL, SQLite, Google BigQuery, Vertica

ETL & APIs: REST APIs, JSON, API Integration, ETL Pipelines

Cloud: AWS, Google Cloud Platform

Version Control & Tools: Git, GitHub, Bitbucket, Azure DevOps, JIRA, Rally, Postman, VS Code, Jupyter Notebook

WORK EXPERIENCE

Mphasis | Chennai | 09/2019 – 03/2025

Data Validation & Automation Engineer

- Validated enterprise-scale datasets between Google BigQuery and Vertica to ensure data consistency.
- Conducted root cause analysis of complex cabin mapping issues and collaborated with Data Engineering teams.
- Supported AWS-to-GCP cloud migration by validating millions of records across environments.
- Automated enterprise data validation workflows using Selenium and Java.
- Reduced production data quality issues through proactive reconciliation and anomaly detection.

Automation & Data Quality Analyst

- Automated validation processes, cutting manual effort by 60%.
- Designed REST API validation frameworks using Postman.
- Integrated automated validation into Azure DevOps CI/CD pipelines.
- Validated backend JSON payloads ensuring data consistency.
- Generated execution reports.

QA Automation Engineer

- Developed Selenium automation frameworks using Java.
- Automated regression testing across enterprise applications.
- Built reusable API validation scripts.

- Integrated user feedback loops with development sprints, resulting in a 15% decrease in critical software defects reported by end-users within the first quarter.

DATA SCIENCE PROJECTS

Project 1: Harvard's Artifacts Collection – ETL, SQL Analytics & Streamlit Showcase

Key Skills: Python, SQL, MySQL, ETL, REST API, Pandas, Streamlit, Data Cleaning

- **Overview:** Developed an end-to-end ETL and analytics platform using the Harvard Art Museum's API to automate data collection, storage, and interactive exploration of museum artifacts.
- **Objective:** Build a scalable data engineering solution to extract, transform, store, and analyze artwork metadata while enabling users to explore collections through an interactive web application.
- **Action:** Designed an automated ETL pipeline using Python and REST APIs, cleaned and normalized JSON data, stored structured data in MySQL, developed optimized SQL queries for analysis, and built an interactive Streamlit dashboard for visualization and search.
- **Outcome:** Successfully automated the complete data ingestion and analytics workflow, improving data accessibility and delivering a scalable platform for artifact exploration and reporting.

Project 2: EcoType – Forest Cover Type Prediction Using Machine Learning

Key Skills: Python, Scikit-learn, Machine Learning, EDA, Feature Engineering, Random Forest, XGBoost, Streamlit

- **Overview:** Developed an end-to-end machine learning solution to classify forest cover types based on environmental and geographical features.
- **Objective:** Predict the dominant forest cover type to support ecological monitoring and land management through accurate multi-class classification.
- **Action:** Performed exploratory data analysis (EDA), data preprocessing, and feature engineering. Trained and evaluated multiple machine learning models, including Logistic Regression, Decision Tree, Random Forest, KNN, and XGBoost, selected the best-performing model, and deployed it using Streamlit.
- **Outcome:** Built a complete machine learning pipeline from data preprocessing to deployment, enabling real-time forest cover prediction. Evaluated model performance using Accuracy-95.4%, Precision, Recall, and F1-score.

Project 3: Aerial Object Classification & Detection using Deep Learning

Key Skills: TensorFlow, Keras, CNN, YOLOv8, OpenCV, Deep Learning, Computer Vision, Streamlit

- **Overview:** Developed a deep learning-based computer vision system to classify birds and drones in aerial images, with optional object detection using YOLOv8.
- **Objective:** Enhance aerial surveillance, airport safety, and wildlife monitoring by accurately identifying birds and drones from aerial imagery.
- **Action:** Built image preprocessing pipelines using OpenCV, developed Convolutional Neural Network (CNN) models with TensorFlow and Keras, integrated YOLOv8 for object detection and localization, and deployed the solution using Streamlit.
- **Outcome:** Delivered an end-to-end deep learning application for aerial image classification-95.8% and object detection- 89.4%, enabling real-time predictions for surveillance and monitoring use cases.

EDUCATION

B.E., (Electronics and Communication Engineering) from Agni college of technology, Chennai, 2017

CERTIFICATIONS

- Master Data Science Program – GUVI
- Selenium & Java Foundation Certification
- Summit Award for Excellence – Mphasis
- SPOT Award – Mphasis